

Experiment 'p_3_8_two_pairs' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_8_two_pairs

Natural language statement: null

Method signature: p_3_8_two_pairs (i: int, j: int, k: int, l: int) returns (s: string)

Ensures

- if (i == j && k == l) || (i == k && j == l) || (i == l && j == k) then s == "two pairs"
else s == "not two pairs"

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_8_two_pairs (i: int, j: int, k: int, l: int) returns (s: string)`

The method should respect the following contract:

`ensures if (i == j && k == 1) || (i == k && j == 1) || (i == 1 && j == k) then s == "two
↪ pairs" else s == "not two pairs"`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn3CoNtQrRuwyUPGEZcuwucD2nXCT

Model: gpt-5.5-2026-04-23

Created at: 1780582766

Finish reason: stop

Usage: 348 tokens in, and 164 tokens out

```
{  
  if (i == j && k == 1) || (i == k && j == 1) || (i == 1 && j == k) {  
    s := "two pairs";  
  } else {  
    s := "not two pairs";  
  }  
}
```

Final Program

```
method p_3_8_two_pairs (i: int, j: int, k: int, l: int) returns (s: string)  
  ensures if (i == j && k == 1) || (i == k && j == 1) || (i == 1 && j == k) then s  
    ↪ == "two pairs" else s == "not two pairs"  
{  
  if (i == j && k == 1) || (i == k && j == 1) || (i == 1 && j == k) {  
    s := "two pairs";  
  } else {  
    s := "not two pairs";  
  }  
}
```

```
}  
}
```

Total Token Usage

Input tokens: 348

Output tokens: 164

Reasoning tokens: 84

Sum of ‘total tokens’: 512

Experiment Timings

Overall Experiment started at 1780582766455, ended at 1780582811043, lasting 44588ms (44.59 seconds)

Iteration #1 started at 1780582766456, ended at 1780582811043, lasting 44587ms (44.59 seconds)